

OpenClaw Plant Care Assistant Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Configuration Tests] in the `Preparation Before Use` directory. If you are only using TUI/WhatsApp, you can skip `04`. This article assumes the basic environment is ready and only covers the plant care project itself.

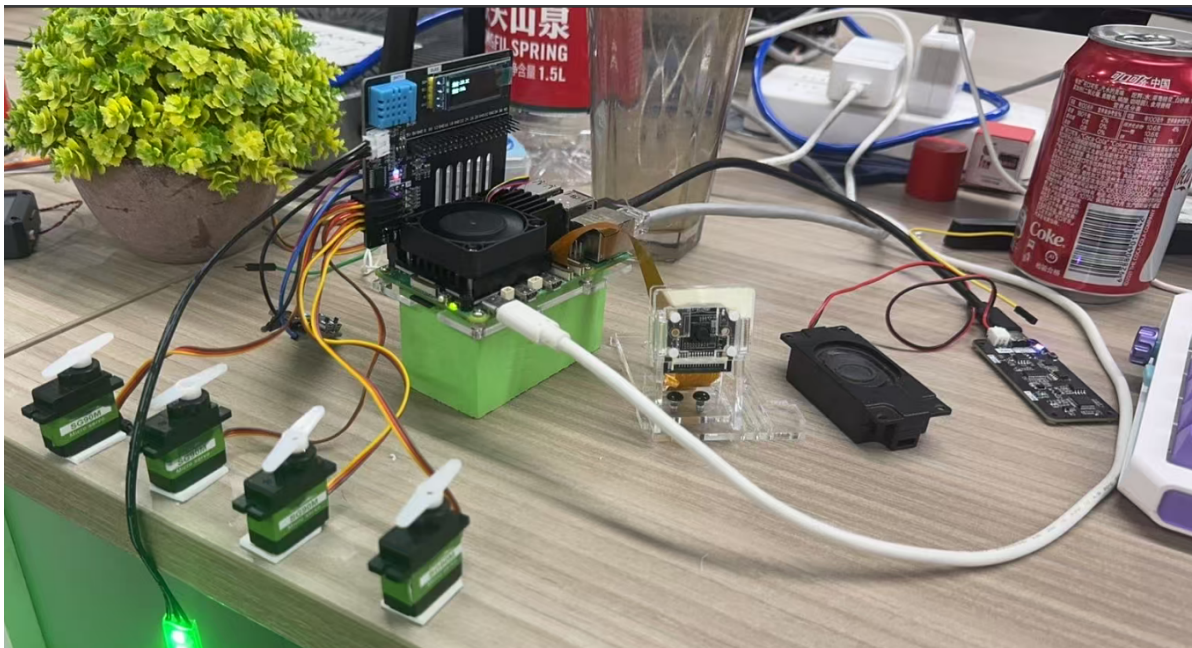
1. Tutorial Objectives

This article demonstrates OpenClaw's multi-peripheral coordination capability through the plant care scenario.

- Understand how to combine temperature and humidity, OLED, RGB, and camera into a plant care assistant
- Communicate with OpenClaw via WhatsApp, TUI, and voice commands

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display screen
- RGB light strip / RGB LED module
- CSI camera
- [Optional] AI voice module + speaker
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial:

- Camera capture command: `/home/pi/hardware_ha1_demo/camera`
- Visual analysis script: `/home/pi/.openclaw/workspace/skills/pi-vision/scripts/csi_describe`

- Voice entry: `/home/pi/voice_to_openclaw/src/main.py`

This project essentially does not rely on a single "big project script" but lets OpenClaw orchestrate multiple basic tools:

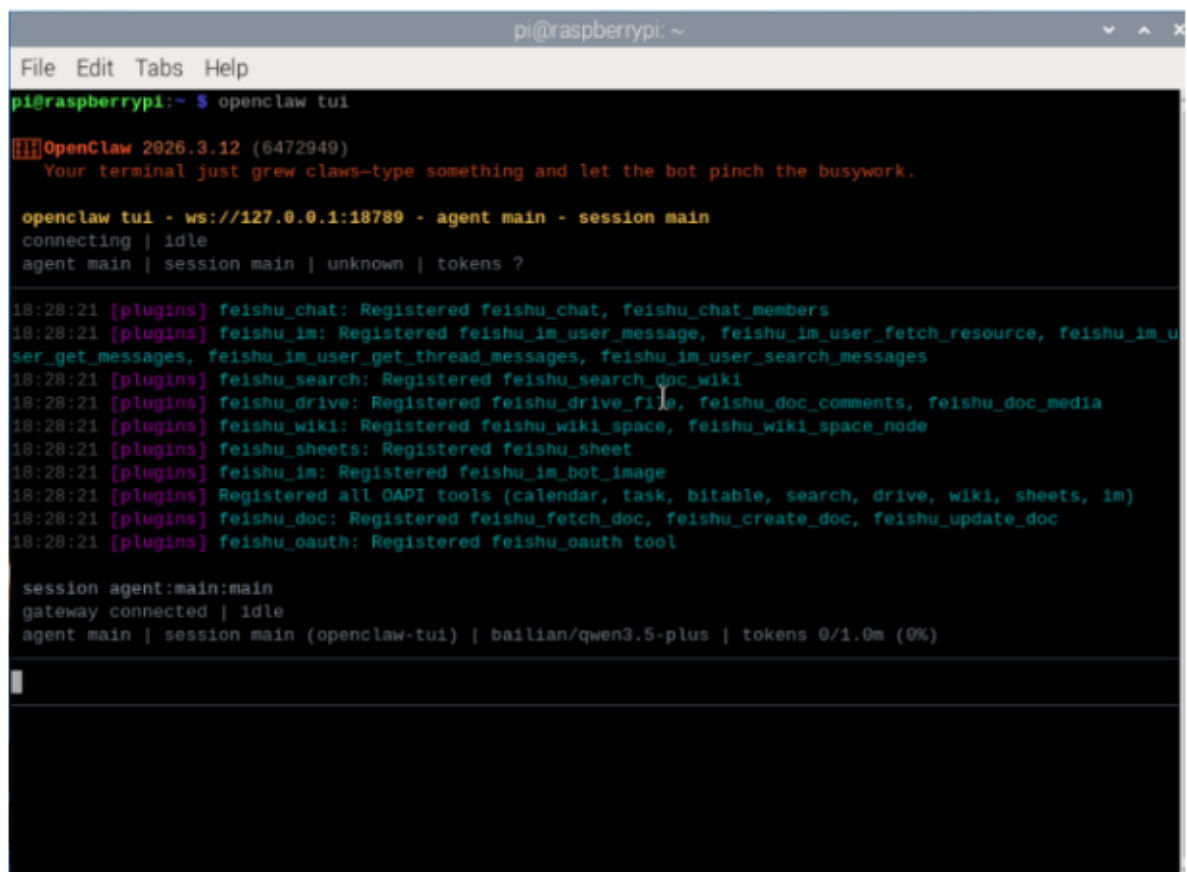
- `dht` is responsible for reading ambient temperature and humidity
- `oled` is responsible for displaying care information
- `rgb` is responsible for status reminders
- `camera` and `csi_describe` are responsible for observing plant appearance

4. Testing Project Basic Capabilities

Before starting this project, please complete an `openclaw tui` basic conversation self-check first; if you haven't done this yet, see [05-Three Dialogue Configuration Tests]. After passing, use TUI to separately verify the basic capabilities in the project.

Input in terminal:

```
openclaw tui
```



```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~ $ openclaw tui  
[[[[OpenClaw 2026.3.12 (6472949)  
Your terminal just grew claws-type something and let the bot pinch the busywork.  
  
openclaw tui - ws://127.0.0.1:18789 - agent main - session main  
connecting | idle  
agent main | session main | unknown | tokens ?  
  
18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members  
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u  
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages  
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki  
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media  
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node  
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet  
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image  
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)  
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc  
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool  
  
session agent:main:main  
gateway connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)  
  
|  
  
|
```

After confirming OpenClaw can dialogue normally, you can try the following types of commands in sequence:

- Read current temperature and humidity
- Display "Plant OK" on OLED
- Change RGB lights to green
- Take a photo for me

If these individual capabilities can all run smoothly, the main chains required for the project are ready.

5. Three Ways to Communicate with OpenClaw

Method	Suitable Scenario	Advantages	Suggested Use
WhatsApp	Remote care, classroom demonstration	Easy to connect, suitable for remote status viewing	Demonstrate "remote natural language care"
TUI	Local debugging, project coordination	Most direct, easiest to troubleshoot	First to get the multi-peripheral chain working
Voice Code	Voice companion, complete AI experience	More like a real assistant experience	As the final demonstration solution

If you have the optional AI voice module + speaker, the TUI and WhatsApp conversations need to broadcast OpenClaw's replies. You can run the following program in the terminal. This is not needed for AI voice module conversations.

- Modify the TTS enable parameter

```
/home/pi/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast replies. After saving, run the program below in the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/pi/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [05-Three Dialogue Configuration Tests]. This section only includes dialogue examples suitable for the plant care project.

After creating the bot successfully, you can remotely inquire about plant status in WhatsApp, for example:

- Take a photo to see the current plant status and perform plant care

20:10



< 2



+86 153 3885 7526 (你)

给自己发消息

Take a photo to see the current plant status and perform plant care

20:09 ✓✓



I can see the plant! It's a small potted plant with vibrant green-yellow leaves. The foliage looks full and healthy. Let me complete the plant care workflow.

20:10 ✓✓

Plant care check completed!

Status:

- 📷 **Visual check:** Plant looks healthy with vibrant green-yellow foliage
- 🌡️ **Temperature:** 29.7°C (85.5°F)
- 💧 **Humidity:** 52%
- 📺 **OLED:** Displaved "Plant:"



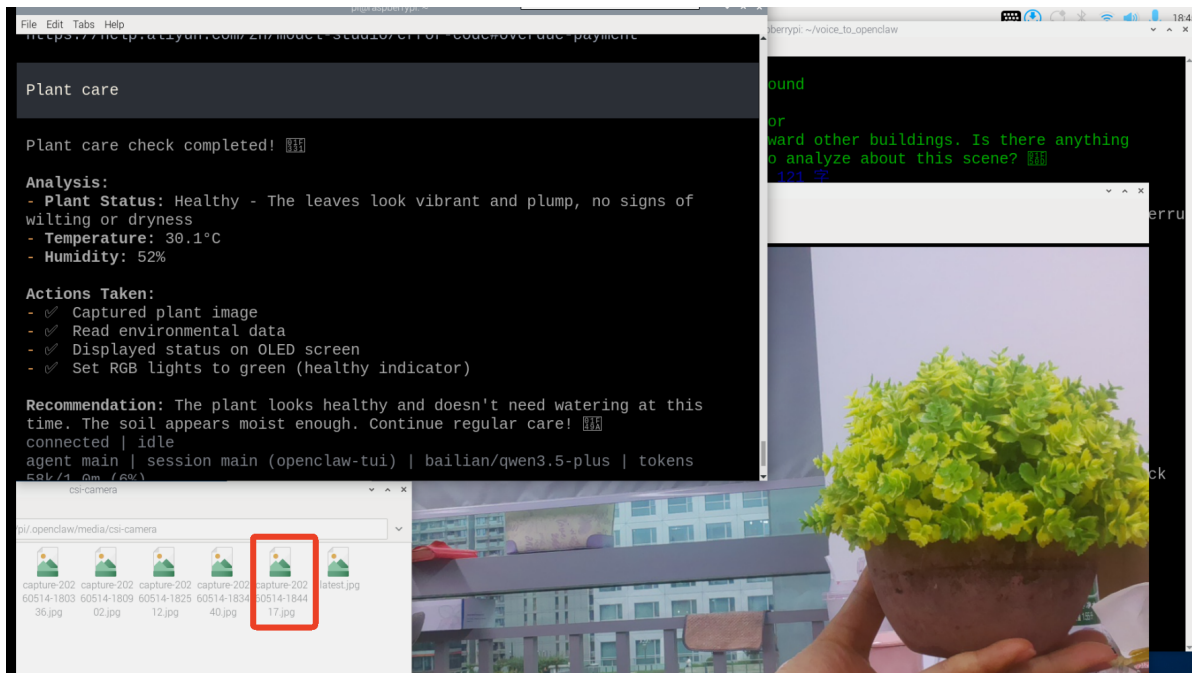
5.2 TUI Solution

If you have already completed the TUI self-check from chapter 0, directly enter terminal text dialogue:

```
openc1aw tui
```

Then you can directly input project-related commands, for example:

- Plant care



5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up flow, please refer directly to [04-AI Voice Interaction Configuration]. After waking up, you can directly speak control content, for example:

- Help me check the plant status and whether it's suitable for watering now


```
pi@raspberrypi: ~/voice_to_openclaw
File Edit Tabs Help
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Me check the plant status and whether it's suitable for watering now.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Let me analyze the plant's condition and check the environment to
           determine if it needs watering.

## 🌱 Plant Status Check
           **Environmental Conditions:**
- **Temperature:** 30.1°C (86.2°F) -
           Quite warm! 🌡️
- **Humidity:** 52% - Moderate

**Plant Analysis:**
           Looking at the captured image, I notice:
- The plant has **bright
           green/yellow foliage** with uniform coloring
- Leaves appear
           **vibrant and upright** (no wilting visible)
- The foliage has an
           **artificial appearance** (plastic-like sheen)

**🚰 Watering
           Recommendation:**
```

6. Project Implementation

Note: All demonstrations below use the `openclaw tui` method for conversation. You can also choose WhatsApp or voice to complete.

6.1 Scenario 1: Environment Reading

Experiment Objective

Read the temperature and humidity around the current plant to complete the most basic environmental perception.

Effect Description

OpenClaw will call DHT to read current ambient data and feed back the temperature and humidity in natural language.

Example Commands

- Read the temperature and humidity near the current plant
- What are the current ambient temperature and humidity
- Help me check if the area around the potted plant is too dry

6.2 Scenario 2: Status Judgment

Experiment Objective

Have OpenClaw give a brief plant care suggestion after getting the ambient data.

Effect Description

Besides directly reporting values, OpenClaw can also summarize "too dry", "too hot", "status normal", etc. in a way closer to daily language.

Example Commands

- Judge the plant status based on current temperature and humidity
- Is it suitable for watering now

6.3 Scenario 3: Audio-Visual Reminders

Experiment Objective

When the environmental status is not ideal, use OLED and RGB to give more intuitive reminders.

Effect Description

For example, light green when the environment is normal, yellow when it's too hot, and red when abnormal; at the same time, display simple prompt information on OLED, so it's not just "can speak" but has clear peripheral feedback.

Example Commands

- Light up green light if the environment is normal
- Light up yellow light and prompt me if it's too hot
- Display "Plant Needs Care" on OLED

Extensible Play

- Link with buzzer for stronger reminders
- Make a desktop care status light
- Add scheduled polling and automatic refresh

6.4 Scenario 4: Photo Observation

Experiment Objective

Use the camera to take a photo of the plant's current state and have OpenClaw give a simple description.

Effect Description

This phase is suitable for observing leaf, flowerpot surface, or overall growth. The current tutorial focuses on "take a look and describe", not automatic disease judgment, so it's more suitable for teaching demonstrations and lightweight auxiliary observation.

Example Commands

- Take a photo of the plant
- Check the leaf status
- Take a photo and describe the current plant image